

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Supplementary Winter Examination – 2023**

**Course: B. Tech.                      Branch : Civil & Allied Engineering                      Semester : IV**

**Subject Name: Water Resources Engineering**

**Subject Code : BTCVC404/BTCIC404/BTCEC404**

**Max Marks: 60**

**Date: 23/1/2024**

**Duration: 3 Hr.**

***Instructions to the Students:***

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

|                                                                                                                                                                                                                                                                                                                                                  | (Level/CO) | Marks     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q.1 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                       |            | <b>12</b> |
| A) An irrigation canal has gross command area of 90000 hectares out of which 80% is culturable irrigable. The intensity of irrigation of Kharif is 40% and for Rabi season is 70%. Find the discharge required at the head of the canal if the duty at its head is 800 hectares/cumec for Kharif season and 1700 hectares/cumec for Rabi season. | <b>CO1</b> | <b>6</b>  |
| B) Write a short note on crop season and crop pattern.                                                                                                                                                                                                                                                                                           | <b>CO1</b> | <b>6</b>  |
| C) Explain briefly about advantages and ill-effects of irrigation.                                                                                                                                                                                                                                                                               | <b>CO1</b> | <b>6</b>  |
| <b>Q.2 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                       |            | <b>12</b> |
| A) Write short note on the selection criterion of site for reservoir.                                                                                                                                                                                                                                                                            | <b>CO2</b> | <b>6</b>  |
| B) Explain the steps followed to calculate of reservoir capacity for a specific yield from mass inflow curve. Explain with proper diagram.                                                                                                                                                                                                       | <b>CO2</b> | <b>6</b>  |
| C) Explain the various classifications of reservoirs.                                                                                                                                                                                                                                                                                            | <b>CO2</b> | <b>6</b>  |
| <b>Q.3 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                       |            | <b>12</b> |
| A) Explain briefly the causes of failure of gravity dam.                                                                                                                                                                                                                                                                                         | <b>CO2</b> | <b>6</b>  |
| B) What is a spillway? What are the various types of spillways?                                                                                                                                                                                                                                                                                  | <b>CO2</b> | <b>6</b>  |
| C) The peak of flood hydrograph due to 3h duration storm in a catchment is 250 cumecs. The total depth of rainfall is 5.6cm. Assuming an average infiltration loss of 0.2 cm/h and a constant base flow of 15 cumecs.                                                                                                                            | <b>CO3</b> | <b>6</b>  |

1. Estimate the peak of the 3h-UH of this catchment.
2. If the area of this catchment is 500 sq.km., determine the base width of the 3h-UH by assuming it to be triangular in shape.

**Q.4 Solve Any Two of the following.** **12**

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|----------------------------------------------------------------------------------------------------|------------|----------|
| A) Enlist the various types of cross drainage works and explain three of them with proper diagram. | <b>CO2</b> | <b>6</b> |
| B) Explain the three methods of base flow separation from flood hydrograph with proper diagram.    | <b>CO3</b> | <b>6</b> |
| C) Why is the lining of the canal necessary?                                                       | <b>CO2</b> | <b>6</b> |

**Q. 5 Solve Any Two of the following.** **12**

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| A) Explain the curative measures of reclamation of water logged land.                                                                                                                                                                                                  | <b>CO1</b> | <b>6</b> |
| B) A catchment area has 7 raingauge stations. In a year the annual rainfall recorded by the gauges are as follows. For a 5% error in the estimation of mean rainfall, calculate the minimum number of additional stations required to be established in the catchment. | <b>CO3</b> | <b>6</b> |

| Raingauge Station | A   | B   | C   | D   | E   | F   | G   |
|-------------------|-----|-----|-----|-----|-----|-----|-----|
| Rainfall (cm)     | 130 | 142 | 118 | 108 | 165 | 102 | 146 |

- |                                               |            |          |
|-----------------------------------------------|------------|----------|
| C) Compare between lift and canal irrigation. | <b>CO2</b> | <b>6</b> |
|-----------------------------------------------|------------|----------|

**\*\*\* End \*\*\***